

Module 8

Z Transform

Exercise Sheet

Section A – Fundamental Concepts

Problem 1

Define the Z-Transform.

- Explain why the Z-Transform is used.
- Compare it with the Discrete-Time Fourier Transform (DTFT).

Problem 2

Explain the Region of Convergence (ROC).

Find the ROC of the following signals:

$$(a) x[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$(b) x[n] = 2^n u[-n - 1]$$

$$(c) x[n] = \left(\frac{1}{2}\right)^{|n|}$$

Problem 3

Determine whether the Z-Transform exists for the following signals.

$$(a) \left(\frac{1}{3}\right)^n u[n]$$

$$(b) 3^n u[n]$$

$$(c) n \left(\frac{1}{2}\right)^n u[n]$$

$$(d) (-0.5)^n u[n]$$

Justify your answers using the ROC.

Problem 4

Find the Z-Transform and ROC of

(a) $\delta[n]$

(b) $u[n]$

(c) $\left(\frac{1}{2}\right)^n u[n]$

(d) $n \left(\frac{1}{3}\right)^n u[n]$

Problem 5

Generate and plot the following discrete-time signals using Python.

- Unit impulse
- Unit step
- Exponential sequence
- Ramp sequence

Comment on their characteristics.

Section B – Properties of the Z-Transform

Problem 6

Verify the Linearity Property.

Find the Z-Transform of

$$2 \left(\frac{1}{2}\right)^n u[n] + 3 \left(\frac{1}{4}\right)^n u[n]$$

using

- analytical calculation
- linearity property
- Python verification

Problem 7

Verify the Time-Shifting Property.

Find the Z-Transform of $x[n - 3]$

where $x[n] = \left(\frac{1}{2}\right)^n u[n]$

Plot both sequences.

Problem 8

Verify the Scaling Property.

Given $x[n] = \left(\frac{1}{2}\right)^n u[n]$

find the Z-Transform of $2^n x[n]$

Explain how the ROC changes.

Problem 9

Verify the Time-Reversal Property.

Find the Z-Transform of $x[-n]$

where $x[n] = \left(\frac{1}{2}\right)^n u[n]$

Problem 10

Use the Initial and Final Value Theorems to determine the initial and steady-state values of

$$(a) X(z) = \frac{z}{z - \frac{1}{2}}$$

$$(b) X(z) = \frac{z}{(z-1)(z-\frac{1}{3})}$$

Verify your answers analytically.

Section C – Programming Problems

Problem 11

Pole-Zero Analysis

Given

$$H(z) = \frac{z-0.5}{(z-0.2)(z-0.8)}$$

- Find poles.
- Find zeros.
- Plot the pole-zero diagram.
- Determine whether the system is stable.

Problem 12

Compute the inverse Z-Transform of

$$X(z) = \frac{z}{z-0.5}$$

Plot the resulting sequence.

Problem 13

Solve the difference equation

$$y[n] - 0.5y[n-1] = x[n]$$

where $x[n] = u[n]$

Plot the output sequence.

Problem 14

Compare the impulse responses of the systems

$$H_1(z) = \frac{z}{z-0.3}$$

and

$$H_2(z) = \frac{z}{z-0.8}$$

Which system has a longer memory? Explain using the pole locations.

